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# Performance of $Bi_2Te_3$-based thermoelectric modules tailored by diffusion barriers

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ABSTRACT

Fifty-pair $Bi_2Te_3$-based thermoelectric (TE) modules were produced by spreading Sn alloy solder on Cu electrodes before heating at 270 °C for 10 min to connect $Bi_2Te_3$-based legs with Cu electrodes. To suppress unexpected reactions between solder and TE materials, a conventional electroless nickel (EN) layer and a novel coating encompassing EN, electroless palladium, and immersion gold (ENEPIG) layers were adopted for $Bi_2Te_3$ materials as diffusion barriers. Influence of different diffusion barriers, namely, EN and ENEPIG, on TE performance of module was ascertained. Interfacial structure at $Bi_2Te_3$/solder/Cu electrode interface revealed that intermetallic compounds were generated owing to interdiffusion among Cu, Sn, and coating elements, and they mostly concentrated at $Bi_2Te_3$/solder and solder/Cu electrode interfaces. Additionally, ENEPIG coating exhibited much better wettability with solder than the EN coating, resulting in fewer defect sites at $Bi_2Te_3$/solder interface. When current flowed through TE modules, the ENEPIG-coated module yielded lower temperatures on both hot and cold sides than the EN-coated module. While the modules had temperature differences between two substrates, output power generated by the ENEPIG-coated module was higher than that of the EN-coated module. In addition, the ENEPIG-coated module showed a smaller internal resistance than the EN-coated module. After heating at 200 °C for 200 h, both modules presented markedly reduced output power, whereas internal resistance was escalated.

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## 1. Introduction

Efforts to research energy harvesting materials cannot be excluded from the tremendous strides humans have made in seeking alternative energy. Energy harvesting materials can generate eco-friendly energy at an atomic scale per se using sources, such as light, heat, and mechanical stress [1], without any external material supply. Among many potential energy harvesting materials, thermoelectric (TE) materials are undeniably promising candidates since they attempt to reversibly convert thermal energy into electricity [2,3]. In addition to heating and cooling intents, TE materials take advantage of waste heat to produce electricity, thereby advancing energy supplying efficiency [4]. Contemporary techniques to produce TE devices have been based on two primary approaches using thin-film and bulk materials. Thin-film TE generators are amenable to applications requiring flexibility, compactness, and portability [5,6]. Moreover, thin-film TE materials exhibit the excellent TE performance [7,8]. Unfortunately, the film thickness is a disadvantageous characteristic for film-based TE devices [9,10]. It is a laborious process to boost the temperature difference between two sides of TE module if the material thickness decreases to a micro-scale. The reduction in thickness also provokes degradation in power generation, as the output power attained by TE generators is proportional to the thickness of TE materials. Meanwhile, bulk TE modules are envisaged for high-performance energy converting devices. However, the difficulties with bulk TE devices include poor mechanical properties, and low TE material performance [10–12].

Bismuth telluride ($Bi_2Te_3$)-based alloys have been studied since the early days of TE materials. These materials can exhibit maximal energy conversion efficiency at a low-temperature range up to 200 °C [13]; hence, they are adopted in applications using human body power and portable cooling appliances [14]. A typical configuration of a commercial $Bi_2Te_3$-based TE module encompasses a consecutive arrangement of n- and p-type bulk legs having a thermally parallel but electrically serial connection with conductive electrodes. In addition to TE parameters, the energy conversion efficiency of TE modules is directly impacted by internal resistance, including electrical resistance of TE specimens and contact

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resistance of TE material/electrode joints [15]. Further, contact resistance can vary due to compositions and reactions at TE material/electrode interface. Consequently, minimizing contact resistance is problematic in production of high-efficiency TE modules.

To produce a bulk $Bi_2Te_3$-based TE module, the connections between TE single legs and conductive electrodes are implemented via soldering with Sn or Sn alloys as a solder [16]. One major hurdle of the soldering method is that Sn element can easily react with Te to yield SnTe intermetallic compound (IMC) during soldering [17]. SnTe IMC presents brittle and porous characteristics, resulting in defect sites at $Bi_2Te_3$/solder interfaces [18]. Therefore, SnTe IMC diminishes bonding strength [19], and escalates contact resistance of TE modules [20]. The deterioration in module performance due to SnTe IMC is so severe that depositing a barrier layer on $Bi_2Te_3$ materials is indispensable to inhibit undesired reactions at $Bi_2Te_3$/solder/electrode interfaces.

Owing to the diffusion barrier, improvement in mechanical strength of TE modules has been realized. Wang et al. [21] ascertained that CoP-coated $Bi_2Te_3$ materials exhibited a bonding strength approximately six times higher than non-coated ones. Lin et al. [22] claimed that NiP coating remarkably improved bonding strength of $Bi_2Te_3$/solder interfaces. Besides enhancement of bonding strength, the coating reduces contact resistance and internal resistance, thereby improving TE performance of the $Bi_2Te_3$-based module [23–25]. Moreover, contact resistance is variable by virtue of dissimilar coatings. Co-sputtered $Bi_2Te_3$ samples presented significantly lower contact resistivity than Ni-sputtered ones [26]. Cheng et al. [20] verified that $Bi_2Te_3$-based modules with Ni/Au multilayers yielded lower internal resistance, exhibiting output power slightly higher than those with a Ni single layer. Nevertheless, no interpretation of such a difference was discussed. Meanwhile, the contact resistance of coated $Bi_2Te_3$ materials is substantially affected by heat treatment. Contact resistivity of CoP/$Bi_2Te_3$ samples was intensified after 15-day annealing due to reaction between CoP layer and $Bi_2Te_3$ materials [24].

Nickel is a conventional diffusion barrier for $Bi_2Te_3$-based TE modules [17,23]. However, due to the low reaction rate with Sn [27], the nickel layer presented neutral wetting characteristics with Sn-based solder [28,29]. Consequently, the reliability of TE modules adopting a nickel layer was degraded when the module underwent heat treatment for several hours [30]. It is essential to seek an alternative layer for $Bi_2Te_3$-based TE modules that exhibits a high reaction rate with Sn without reacting with $Bi_2Te_3$. Additionally, one study claimed that gold was the metallic element exhibiting the fastest diffusion rate into Sn-based solder [27]. It was anticipated that adding a gold layer on the nickel would enhance the wettability with solder for the coated TE material [20]. In this study, a novel diffusion barrier comprising electroless nickel (EN), electroless palladium (EP), and immersion gold (IG) (ENEPIG) multilayers was investigated for $Bi_2Te_3$-based TE modules. Using electroless deposition method, an IG layer could be directly deposited on an EN layer. However, the black pads or pinholes caused by the corrosion of EN during immersion process deteriorate the mechanical strength and reliability of solder joints [31,32]. Therefore, supplementing a coating, such as palladium, on EN before gold immersion was commonly preferred to avoid the defect formation. The influence of ENEPIG coating on bonding strength of $Bi_2Te_3$/solder interface was ascertained by comparing it with an EN layer [33]. Recently, research has explored TE performance of $Bi_2Te_3$-based modules with EN and ENEPIG deposits. First, difference in interface structure between EN- and ENEPIG-coated $Bi_2Te_3$/solder joints was examined. Afterward, the cooling performance of modules was validated by recording temperature on both hot and cold sides when current flowed through modules. For power generation performance, output power could be evaluated by applying a temperature difference between two sides of the module. Simultaneously, internal and contact resistances were determined. Variation in contact resistance between modules relating to EN and ENEPIG will be explained on the basis of the difference in interface structure. In addition, the power generating performance of modules with EN and ENEPIG layers after heating at 200 °C for 200 h was characterized.

## 2. Material and methods

The starting materials were Bi, Te, and Sb metallic powders (99.999% purity, Materion Co. Ltd, USA). The initial powders were mixed to stoichiometric ratios of $Bi_{0.5}Sb_{1.5}Te_3$ and $Bi_{1.7}Sb_{0.3}Te_3$ for p- and n-type, respectively, and mechanically alloyed via a high-energy ball milling process using a pulverisette (Planetary Mono Mill Pulverisette, Fritsch, Germany) for 2 h at 450 rpm. The bulk materials were sintered by spark plasma sintering (SPS, Sumitomo, Dr. Sinterlab, Japan) at 350 °C for 10 min under vacuum with a 60-MPa uniaxial pressure. The densified samples underwent sandblasting using alumina particles to increase surface roughness. Prior to electroless deposition, sample surfaces were activated in a palladium catalyst solution (ATOTECH, Adhemax Activator SF, Germany) at 40 °C for 3 min. At the outset of electroless plating, the bulk disks were immersed in a commercial nickel phosphorus electroless solution (MPN-5, KPM Tech Co. Ltd, Korea) at 80 °C for 15 min. Subsequently, the EP layer could be obtained on EN coating by palladium phosphorus electroless deposition at 55 °C for 6 min. The last step of the plating process was gold immersion at 90 °C for 30 min. Coatings retained thicknesses in the ranges of ~3–4 μm, ~0.1–0.2 μm, and ~0.1–0.2 μm for the EN, EP, and IG layers, respectively. After electroless deposition, $Bi_2Te_3$-based samples were cut into cube specimens with 3-mm edge length by wire cutting. To fabricate fifty-pair TE modules, a stacked structure including alumina substrate/Cu electrodes/solder paste/$Bi_2Te_3$ specimens/solder paste/Cu electrodes/alumina substrate was assembled after spreading Sn-3%Ag-0.5%Cu (SAC305) solder paste on Cu electrodes. Such structure was heated at 270 °C for 10 min to connect TE specimens to Cu electrodes. Finally, TE modules were annealed at 200 °C for 200 h.

$Bi_2Te_3$/solder/Cu electrode interface was examined for its microstructure and chemical composition via scanning electron microscopy (SEM) accompanied with energy dispersive spectroscopy (EDS) (SU8220, Hitachi, Japan). To explore the efficiency of TE modules as coolers, temperature on both hot and cold substrate sides was recorded using a thermometer while current flowed through TE modules. Current intensity varied in the range of 0.5–3.5 A. To validate the performance of TE modules as generators, output power of TE modules was measured by applying a temperature gradient. Cold temperature was fixed at 25 °C, and hot temperature was varied from 50 °C to 175 °C in a 25 °C step. Average internal resistance $R_i$ (Ω) and contact resistance $R_c$ (Ω) could be, respectively, calculated as follows [15]:

$$R_i = \frac{(U_{OC} - U_L) \times R_L}{U_L} \quad (1)$$

$$R_c = R_i - (R_P + R_N) = R_i - n \times (\rho_P + \rho_N) \times \frac{l}{S} \quad (2)$$

where $U_{OC}$ is the open-circuit voltage (OCV) (V); $U_L$ is the load voltage (V); $R_P$ and $R_N$ represent the resistances of p- and n-type legs (Ω), respectively; $R_L$ is the load resistance (Ω); $\rho_P$ and $\rho_N$ signify resistivities of p- and n-type legs (Ω.m), respectively; $l$ and $S$ denote height (m) and area ($m^2$) of TE specimens, respectively; and $n$ is the number of TE pairs. The resistivity of TE legs with a dimension of $3 \times 3 \times 10\ mm^3$ could be determined using a four-point method (BS1, BlueSys, South Korea) in a helium atmosphere.

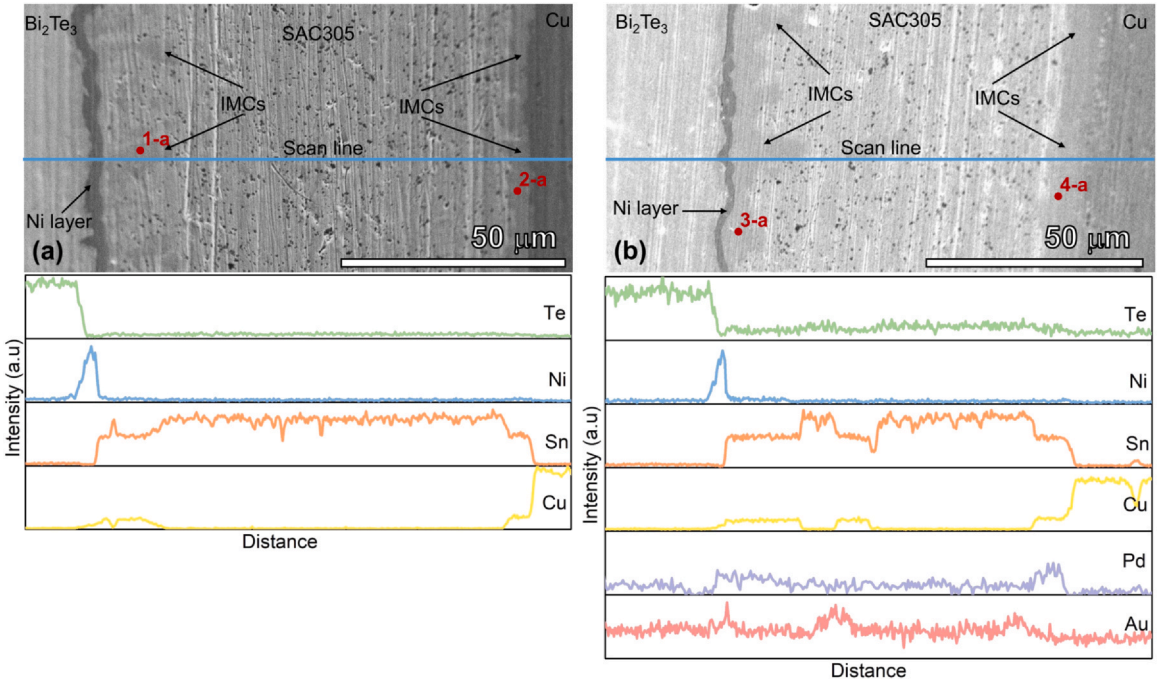



**Fig. 1.** Scanning electron microscopy images and energy dispersive X-ray spectroscopy line-scan analyses across $Bi_2Te_3$/solder interfaces with (a) electroless nickel and (b) electroless nickel, electroless palladium, and immersion gold.

## 3. Results and discussion

### 3.1. Analysis of interface between coated $Bi_2Te_3$ materials and solder

During soldering, heat accompanied with the molten state of solder, facilitates interdiffusion in the solder region. An effective diffusion barrier has a minimal diffusion rate into $Bi_2Te_3$-based alloys; however, it may partially react with metallic elements in molten solder to generate IMCs [16,17,24]. Chemical element composition at $Bi_2Te_3$/solder/Cu electrode interface was characterized via line-scan analysis (Fig. 1) and point analysis (Table 1). Given an element concentration along a scan line, IMCs were determined mostly at $Bi_2Te_3$/solder and solder/Cu electrode interfaces, not in the solder area, which was attributable to heterogeneous nucleation that prevails in liquid–solid phase transformation owing to low activation energy rather than homogeneous nucleation [34]. Additionally, concentration gradient was a driving force for element migration between $Bi_2Te_3$/solder and solder/Cu electrode interfaces. Specifically, Cu element diffused from Cu electrode and solder alloy, which had high Cu concentration, to $Bi_2Te_3$/solder interface, which presented less Cu concentration, in both EN and ENEPIG cases. Meanwhile, Cu diffusion flux into Sn had a higher rate than that of Ni [27], thereby depressing Ni diffusion flux into Sn. Therefore, small Ni content was detected in IMCs at solely $Bi_2Te_3$/solder interface (points 1-a and 3-a in Table 1). For ENEPIG, the complete dissolution of Au element into Sn exposed Pd element to molten solder. Moreover, as Pd diffusion rate into Sn was higher than that of Ni [35–37], Pd could react with Cu and Sn elements and migrate through Sn region to yield IMCs at both $Bi_2Te_3$/solder and solder/Cu electrode interfaces (Fig. 1b, points 3-a and 4-a in Table 1).

**Table 1**
Atom percent (%) of chemical elements in intermetallic compounds.

| Points | Atom percent (%) | | | | Suggested phases |
|---|---|---|---|---|---|
| | Sn | Cu | Pd | Ni | |
| 1-a | 49.37 | 47.96 | – | 2.67 | $(Cu,Ni)_6Sn_5$, Sn |
| 2-a | 41.56 | 58.44 | – | – | $Cu_6Sn_5$, $Cu_3Sn$ |
| 3-a | 59.87 | 32.06 | 3.98 | 4.09 | $(Cu,Ni,Pd)_6Sn_5$, Sn |
| 4-a | 40.54 | 56.66 | 2.80 | – | $(Cu,Pd)_6Sn_5$, $Cu_3Sn$ |
| 1-b | 43.70 | 52.20 | – | 4.09 | $(Cu,Ni)_6Sn_5$ |
| 2-b | 35.84 | 64.16 | – | – | $Cu_6Sn_5$, $Cu_3Sn$ |
| 3-b | 42.69 | 53.70 | 0.49 | 3.12 | $(Cu,Ni,Pd)_6Sn_5$ |
| 4-b | 39.81 | 60.19 | 0.00 | – | $Cu_6Sn_5$, $Cu_3Sn$ |

The phase composition of IMCs was estimated using atom percent of elements attained by point analysis (Table 1). According to Cu–Sn phase diagram [38], Sn dominated at $Bi_2Te_3$/solder interface, and it reacted with Cu to form $Cu_6Sn_5$ and Sn. In addition, it was supposed that Ni and Pd atoms with the exiguous amount occupied Cu atoms in $Cu_6Sn_5$ lattice [39–42]. As a result, IMCs at $Bi_2Te_3$/solder interface encompassed $(Cu,Ni)_6Sn_5$ and Sn for EN, whereas Pd and Ni tended to amalgamate with $Cu_6Sn_5$ to generate $(Cu,Ni,Pd)_6Sn_5$ in IMCs relating to ENEPIG. Otherwise, Cu element prevailed at solder/Cu electrode interface, so it incorporated with $Cu_6Sn_5$ to form $Cu_3Sn$ [43]. Consequently, IMCs at solder/Cu electrode interface involving EN included $Cu_6Sn_5$ and $Cu_3Sn$, whereas IMCs with ENEPIG contained $(Cu,Pd)_6Sn_5$ and $Cu_3Sn$.

After the samples were heat-treated at 200 °C for 200 h, chemical composition at $Bi_2Te_3$/solder/Cu electrode interface was also analyzed via line-scan and point analyses (Fig. 2 and Table 1). IMCs distributed thicker and denser at both $Bi_2Te_3$/solder and solder/Cu electrode interfaces than compounds with no heating, regardless of EN and ENEPIG. According to the atom percent of elements (Table 1), Cu quantity increased in IMCs at both sides of the solder region. Therefore, IMC composition at $Bi_2Te_3$/solder interface with EN coating included only $(Cu,Ni)_6Sn_5$. Interestingly, Pd content in IMCs relating to ENEPIG declined, and no existence of Pd element was detected in IMC at solder/Cu electrode interface. Indeed, the element concentration along the scan line (Fig. 2b) showed that most Pd dissolved into solder alloy after annealing. Consequently, $(Cu,Ni,Pd)_6Sn_5$ IMC was yielded with an infinitesimal Pd amount of 0.49 at% at $Bi_2Te_3$/solder interface with the ENEPIG layer. The IMC compositions at solder/Cu electrode interfaces involving EN and ENEPIG were analogous, comprising $Cu_6Sn_5$ and $Cu_3Sn$.

As the soldering process took advantage of the molten state of solder to bond $Bi_2Te_3$ materials and Cu electrodes, the wettability of both $Bi_2Te_3$ and Cu surfaces with solder played a central role in

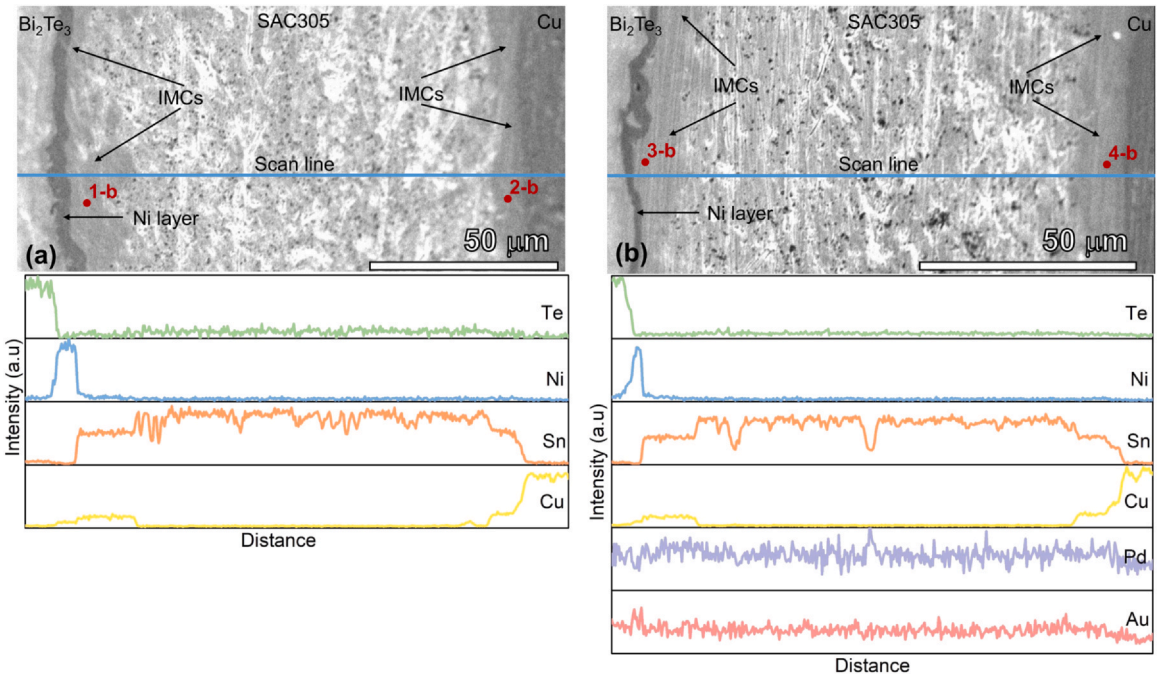



**Fig. 2.** Scanning electron microscopy images and energy dispersive X-ray spectroscopy line-scan analyses across 200-h-heated $Bi_2Te_3$/solder interfaces with (a) electroless nickel and (b) electroless nickel, electroless palladium, and immersion gold.

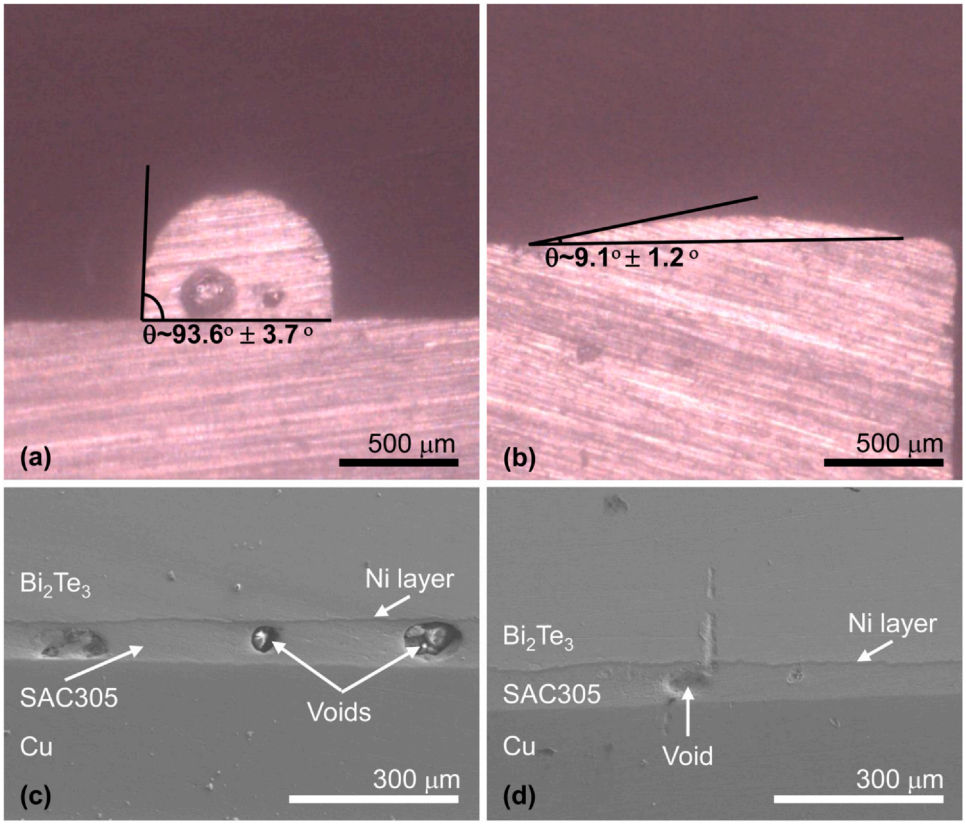



**Fig. 3.** Contact angles of melted solder with (a) electroless nickel (EN)- and (b) electroless nickel, electroless palladium, and immersion gold (ENEPIG)-coated $Bi_2Te_3$-based surfaces; microstructure of $Bi_2Te_3$/solder interface with (c) EN and (d) ENEPIG.

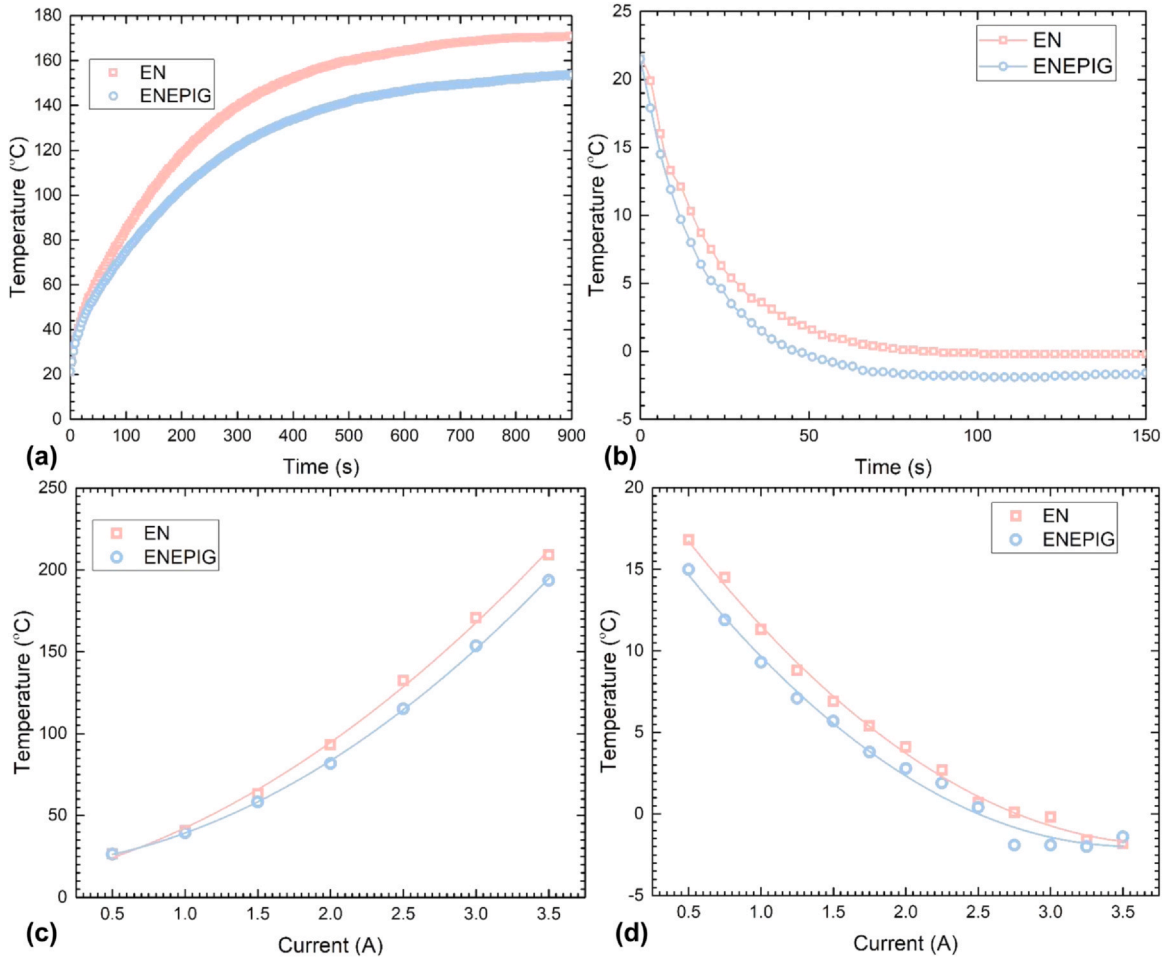



**Fig. 4.** Temperature over time on (a) hot side and (b) cold side at 3-A current intensity; stable temperature as a function of current on (c) hot side and (d) cold side.

interfacial adhesion [44,45]. The solder spread on identical Cu surfaces; hence, only the contact angles of solder ball with coated $Bi_2Te_3$ materials after heating at 270 °C for 10 min was examined under side-view observation (Fig. 3a and b). The mean values of contact angles attained by 10 repeated measurements were 93.6° and 9.1° for EN and ENEPIG, respectively. As depicted in Fig. 3a, EN-coated $Bi_2Te_3$ surfaces introduced a right contact angle, representing neutral wetting characteristics with solder. By contrast, the solder ball widely spread on ENEPIG-plated $Bi_2Te_3$ materials, and the contact angle of solder with ENEPIG surface was immensely low, which indicated that the ENEPIG layer showed outstanding wettability with solder. Such high wettability was attributed to the total diffusion of Au into solder alloy (Fig. 1b). In specific, Au dissolution into Sn-based alloys decreased solder surface tension and increased ENEPIG surface energy, thereby ameliorating the wettability of ENEPIG-coated $Bi_2Te_3$ with solder. Owing to the slightly poor wettability of EN coating with solder, voids at interface obtained by EN film outnumbered the ones at interface achieved by ENEPIG layer (Fig. 3c and d).

### *3.2. Influence of coating on performance of $Bi_2Te_3$-based thermoelectric modules*

While current flowed through TE modules, a temperature gradient was generated between two sides of the module according to Peltier effect [46]. However, the complexity of cooling efficiency achieved by TE modules stems from the integration of Peltier, Thomson, and Joule heats in the heat transfer characteristics of the module [47]. Moreover, the heat emitted on the hot side with high energy can easily transfer to the cold side with low energy due to heat absorption, thereby reducing cooling performance. To evaluate the final cooling performance of TE cooler, the temperatures on the hot and cold sides were separately measured. Specifically, the module was hung on a hook isolating it from the vicinity when the temperature on the hot side was being measured. The hot side was placed on a heat sink to inhibit heat transfer from the hot to cold sides, while the temperature on the cold side was being measured. Both the hot and cold temperatures yielded by various current intensities are presented in Fig. 4. At a fixed current intensity, e.g., 3 A (Fig. 4a and b), the initial temperature was fixed in the range of 21–22 °C for both the hot and cold sides. On the hot side, the temperature gradually increased to the maximum value, in approximately 800 s, and subsequently retained such a value. The slope of the line relating to ENEPIG being lower than that relating to EN indicated that ENEPIG-coated module rejected heat at a slower rate than EN-coated ones. Furthermore, the temperature obtained by ENEPIG-coated module had a lower maximum value than that with the EN coating. For the cold side, the temperature reached the minimum value after approximately 60 s, with the cooling rate for ENEPIG higher than that for EN. ENEPIG-coated module could obtain

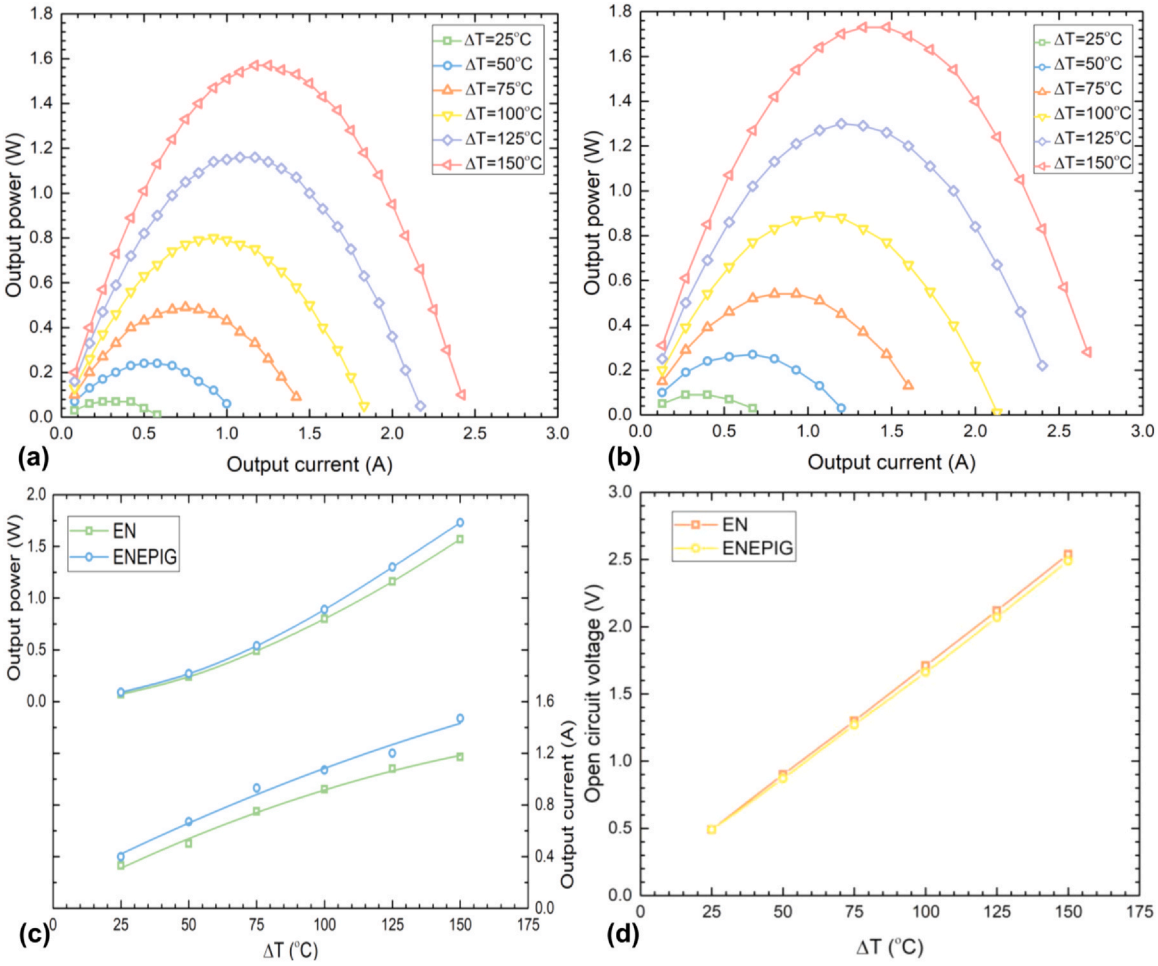



**Fig. 5.** Output power of $Bi_2Te_3$-based thermoelectric modules with (a) electroless nickel and (b) electroless nickel, electroless palladium, and immersion gold induced by various temperature gradients; (c) maximum output power with corresponding output current and (d) open-circuit voltage as a function of temperature gradient.

a lower minimum temperature than EN-coated ones. Alongside the increase in current intensity, the overall tendency was the increase and decrease in constant values of temperature at the hot and cold sides, respectively (Fig. 4c and d). Although the temperature on the hot side was comparable at starting current intensity of 0.5 A for both EN and ENEPIG, EN-coated module always yielded a temperature higher than ENEPIG-coated one when current intensity exceeded 0.5 A. Meanwhile, the modules attained by ENEPIG showed a lower temperature on the cold side than those obtained by EN at most current intensities. Both modules exhibited an identical lowest point of cold temperature, namely, approximately −2 °C. ENEPIG-coated module could generate the coldest temperature at greater than 2.75 A current intensity, whereas EN-coated module required a higher current intensity of greater than 3.25 A to attain the lowest temperature.

According to Seebeck effect, if a temperature difference is applied between two sides of a module, an electromotive force will be formed, generating a current flow in the module [46]. By varying load resistance of a resistor associated with TE module to form a series circuit, output power yielded by TE module could be determined. Fig. 5a and b describe output power attained by EN- and ENEPIG-coated modules, respectively, as a function of load current. Apparently, power always increased with the increase in the temperature gradient. Power generated by the module with ENEPIG was higher than that with EN, regardless of the temperature gradient. Maximum power relating to ENEPIG was slightly greater than that relating to EN when $\Delta T$ was below 100 °C and this difference increased when $\Delta T$ exceeded 100 °C (Fig. 5c). Moreover, ENEPIG-coated module showed an output current higher than EN-coated one with any $\Delta T$. In addition to identification as the product of load voltage ($U_L$) and load current ($I_L$), the output power can be defined in terms of OCV ($U_{OC}$) and internal resistance ($R_i$) as follows [15]:

$$P_{out} = U_L \times I_L = \frac{U_{OC}^2}{4R_i} \quad (3)$$

The OCVs exhibited no appreciable variation in EN- and ENEPIG-coated modules (Fig. 5d).

The internal resistance of modules obtained during the measurement of power-generating performance was evaluated using Eq. (1), which is depicted as a function of $\Delta T$ in Fig. 6a. Internal resistance marginally escalated alongside the increase in $\Delta T$ for EN and ENEPIG cases. Moreover, ENEPIG-coated module always presented a lower internal resistance than EN-coated one with any $\Delta T$. Fig. 6b depicts the resistivities of n- and p-type legs obtained via the four-point probe method. Apparently, resistivity slightly increased with an increase in $\Delta T$. Using Eq. (2), the contact resistance was determined (Fig. 6c). For EN and ENEPIG, the contact resistance was

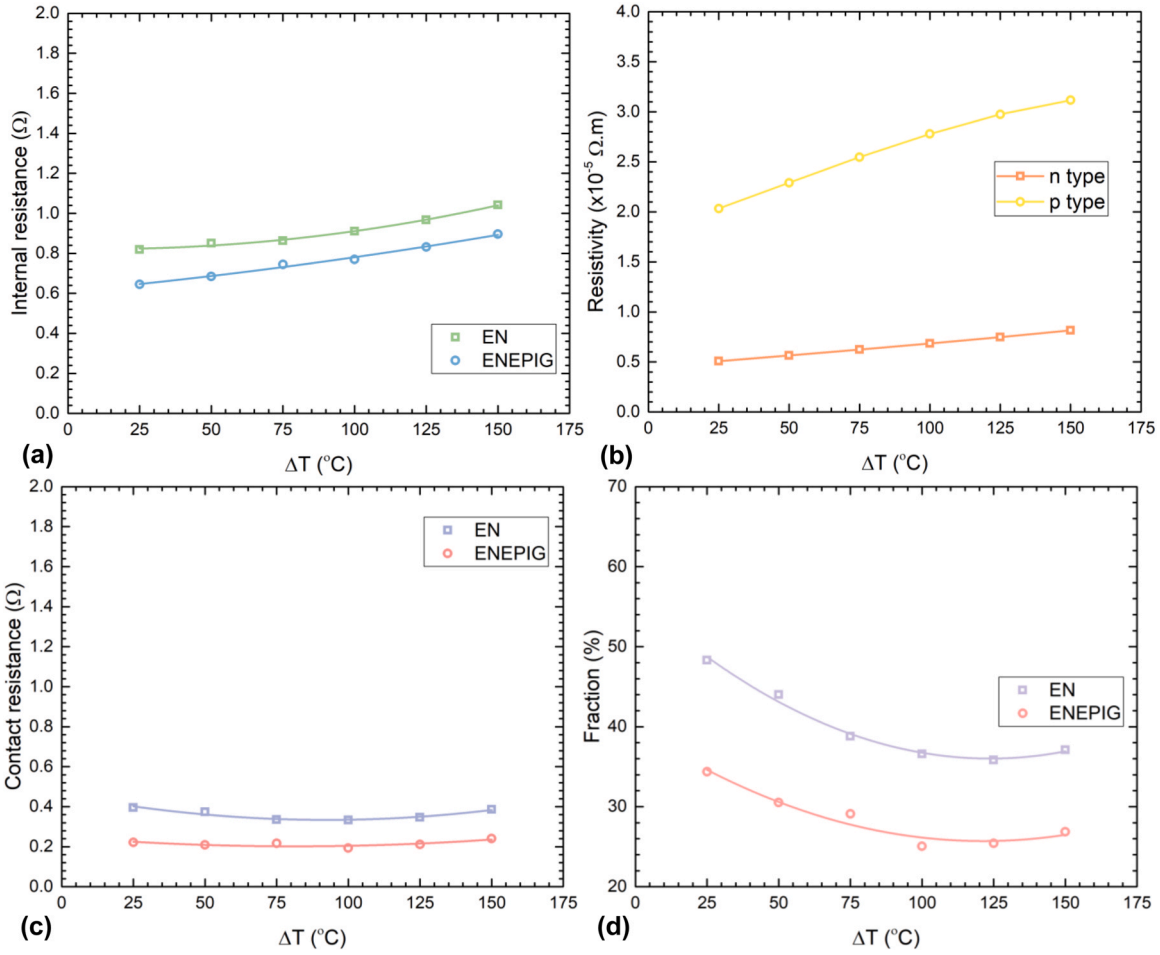



**Fig. 6.** (a) Internal resistance of $Bi_2Te_3$-based modules with electroless nickel (EN) and electroless nickel, electroless palladium, and immersion gold (ENEPIG) coatings, (b) resistivity of n- and p-type $Bi_2Te_3$-based legs, (c) contact resistance, and (d) ratio of contact resistance to the internal resistance of $Bi_2Te_3$-based modules with EN and ENEPIG coatings.

almost stable with an increase in $\Delta T$. The contact resistance achieved by ENEPIG was lower than that achieved by EN regardless of $\Delta T$. The ratio of contact resistance to internal resistance is shown in Fig. 6d. The contact resistance involving EN accounted for a higher proportion of internal resistance than that involving ENEPIG; these proportions tended to reduce when $\Delta T$ increased. From Eq. (3), the output power was influenced by OCV and internal resistance. Despite analogy in OCV between EN- and ENEPIG-coated modules, the internal resistance of EN-coated module was higher than that of ENEPIG-coated module, thereby leading to output power attained by EN less than that by ENEPIG.

One possibility affecting the internal resistance of the TE modules is an interface barrier of a metal-semiconductor heterojunction induced by metallic coating/$Bi_2Te_3$-based alloy interface. Nevertheless, in both the cases of EN and ENEPIG, the metallic layers adjacent to $Bi_2Te_3$ materials were identical, i.e. the nickel layer (Fig. 1). As a result, the influence of interfacial barrier of metal-semiconductor heterojunction on the internal resistance of the TE modules was supposedly comparable between EN- and ENEPIG-coated modules. Therefore, the difference in the internal resistance of EN- and ENEPIG-coated modules directly stemmed from the variation in contact resistance. The variation in contact resistance could be explained by the structure of $Bi_2Te_3$/solder/Cu electrode interface, as discussed in Subsection 3.1. Pd and Au elements presented in $Bi_2Te_3$/solder/Cu interface involving ENEPIG as impurities. According to Matthiessen's rule [48], the presence of impurities in metallic compounds results in scattering charge carriers, thereby intensifying resistivity. Despite the increase in resistance due to impurities, the better wettability of ENEPIG with solder resulted in fewer defect sites, thereby reducing electrical contact resistance more than those of EN. The lower ENEPIG-related contact resistance than EN-related ones verified that the impact of defect sites on contact resistance prevailed rather than that of impurities.

Such a difference in internal resistance might be the cause of the variation in temperature exhibited on the hot and cold sides of EN- and ENEPIG-coated modules (Fig. 4c and d). Specifically, an addition to internal resistance caused an additional Joule heat for heat transfer in the modules. Consequently, EN-coated module showed a higher temperature on both the hot and cold sides than ENEPIG-coated module due to its greater internal resistance. In addition, Joule heat (J) is expressed as the product of current intensity (I) squared with resistance (R) and time (t) as follows: $J = I^2Rt$. Apparently, Joule heat increases more rapid with the increase in current intensity than ones with the increase in resistance. The identical temperatures on the cold sides of both EN and ENEPIG modules at current intensities over 3.25 A verified that a large current intensity

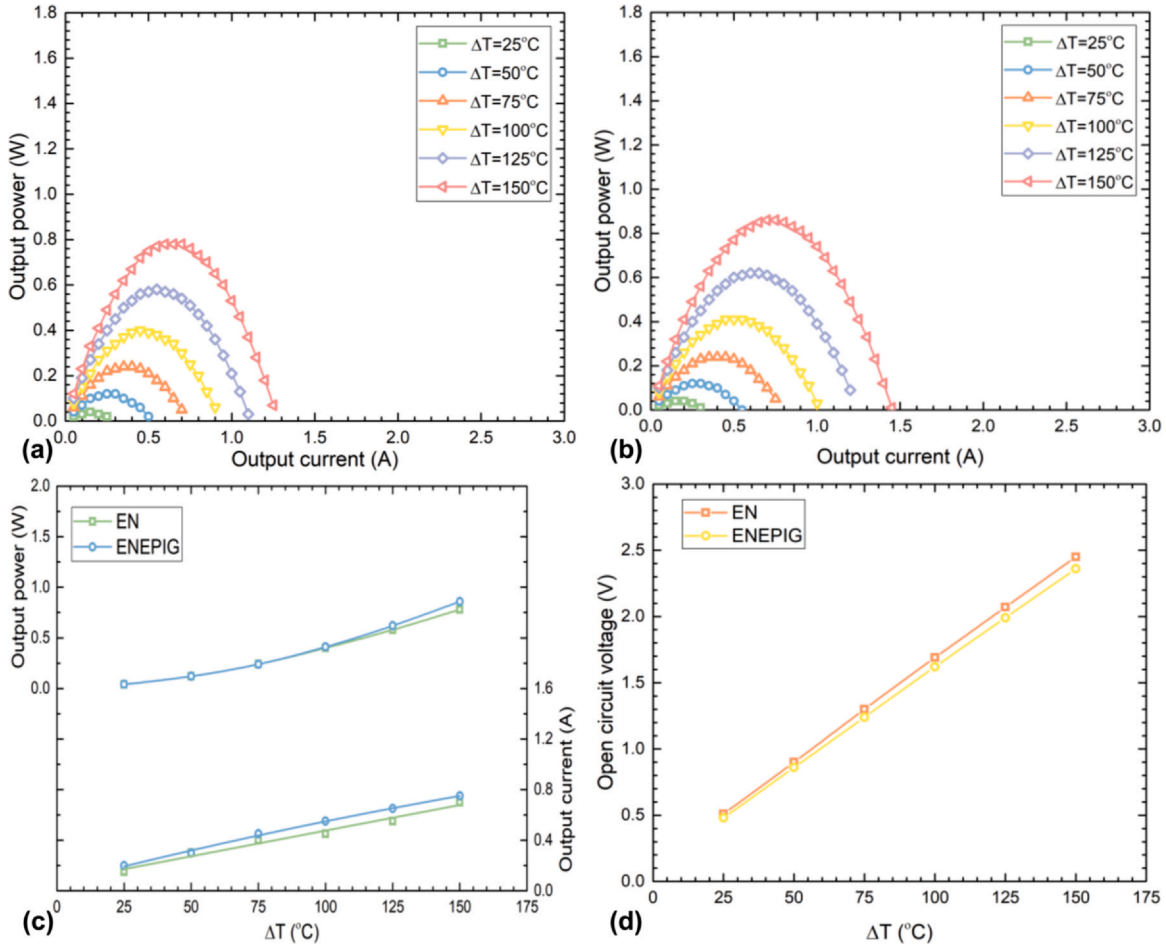



**Fig. 7.** Output power of 200-h-heated $Bi_2Te_3$-based thermoelectric modules with (a) electroless nickel and (b) electroless nickel, electroless palladium, and immersion gold induced by various temperature gradients; (c) maximum output power with corresponding output current and (d) open-circuit voltage as a function of temperature gradient.

generated a high Joule heat amount, diminishing the difference in cooling performance caused by various internal resistances of EN- and ENEPIG-coated TE modules.

### *3.3. Influence of heat treatment on performance of $Bi_2Te_3$-based thermoelectric modules*

Fig. 7a and b show the output power of EN- and ENEPIG-coated modules after annealing at 200 °C for 200 h. Overall, the output power yielded by both modules sharply declined after heat treatment. The maximum powers obtained by EN and ENEPIG were comparable at $\Delta T$ up to 100 °C; however, ENEPIG-coated module could achieve a slightly higher maximum power at $\Delta T$ over 100 °C (Fig. 7c). In addition, the output current corresponding to the maximum value of power showed a minimal difference between EN- and ENEPIG-coated modules. Fig. 7d represents OCV induced by EN- and ENEPIG-coated modules. Results indicated that the OCV of ENEPIG-coated module was marginally lower than that of EN-coated one.

Fig. 8 a presents the internal resistance of modules experiencing heat treatment at 200 °C for 200 h. After heating, the internal resistance of modules markedly intensified compared with that without heating regardless of coating, and it increased slightly when $\Delta T$ increased. ENEPIG-coated module still maintained an internal resistance lower than EN-coated one with any $\Delta T$ after annealing. Although the internal resistance yielded by the module with ENEPIG was remarkably lower than that with EN, the difference in output power between EN- and ENEPIG-coated modules was reduced, which stemmed from the OCV of ENEPIG-coated module slightly being lower than that of EN-coated module. By virtue of the exponent 2 of $U_{OC}$ in Eq. (3), the minimal difference in OCV substantially reduced variation in output power between EN- and ENEPIG-coated modules. The resistivities of n- and p-type legs with heating were mildly higher than those without heating, and these values tended to slightly increase with an increase in $\Delta T$ (Fig. 8b). Fig. 8c indicates that contact resistance with heating appreciably intensified compared with that with no heating. The annealed module involving ENEPIG exhibited a contact resistance that was lower than that involving EN. In addition, the contact resistance of both modules with EN and ENEPIG insignificantly changed with an increase in $\Delta T$. Meanwhile, contact resistance accounted for a vast proportion of internal resistance regardless of coating (Fig. 8d), namely, up to 70% and more than 60% for EN and ENEPIG, respectively, and such proportions had a reducing tendency with increasing $\Delta T$. These results suggested that intensifying internal resistance after heating was mostly attributable to contact resistance. The increase in contact resistance was due to the existence of microcracks at $Bi_2Te_3$/solder interface after annealing [49]. Given the dominance of contact resistance, it could be concluded that the thermal stability of the module after experiencing heat treatment mainly relied on interfacial stability between $Bi_2Te_3$ materials and Cu electrodes.

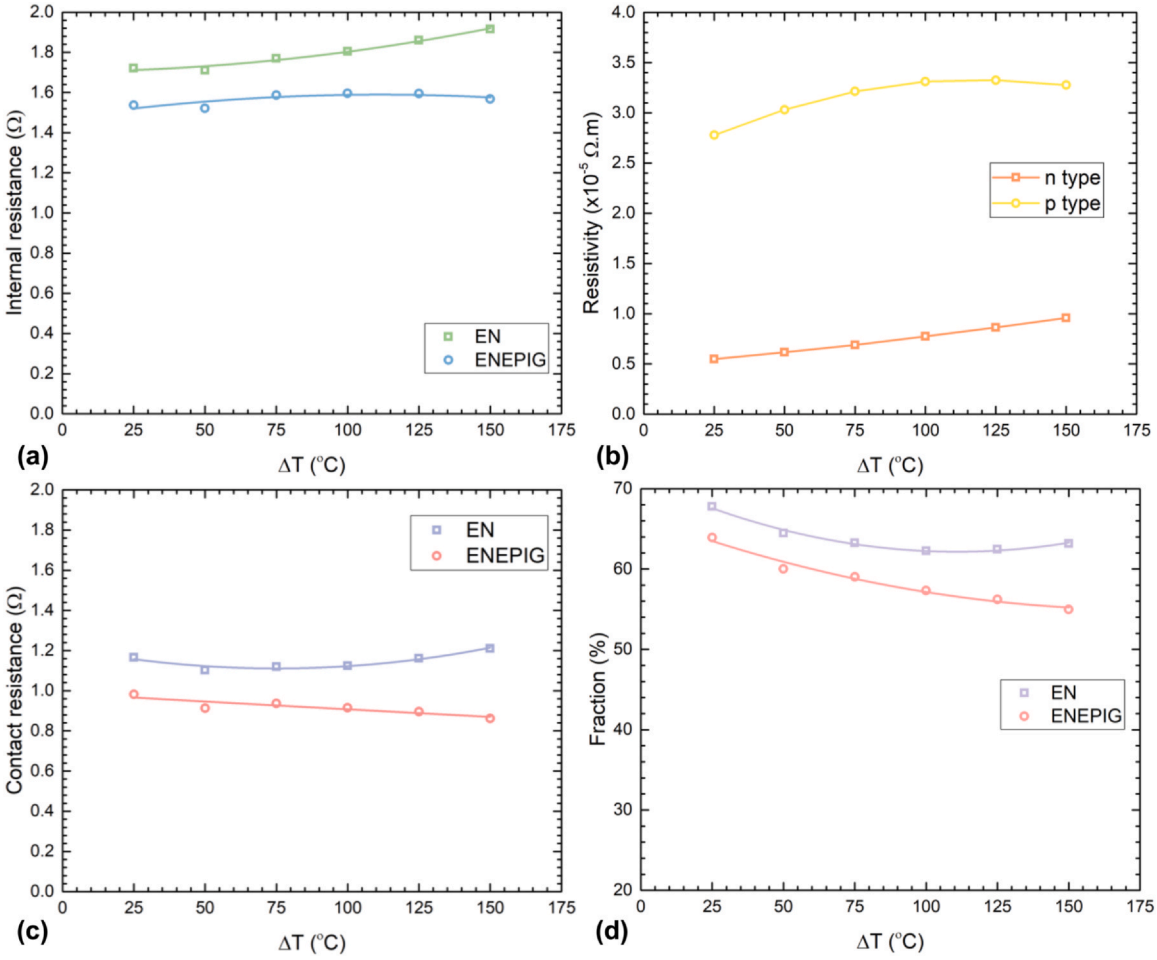



**Fig. 8.** (a) Internal resistance of 200 h-heated $Bi_2Te_3$-based modules with electroless nickel (EN) and electroless nickel, electroless palladium, and immersion gold (ENEPIG) coating, (b) resistivity of 200 h-heated n- and p-type $Bi_2Te_3$-based legs, (c) contact resistance, and (d) ratio of contact resistance to the internal resistance of 200 h-heated $Bi_2Te_3$-based modules with EN and ENEPIG coating.

## 4. Conclusions

Fifty-pair $Bi_2Te_3$-based TE modules were produced using EN and ENEPIG as diffusion barriers. Interfacial analysis verified that IMCs were formed at $Bi_2Te_3$/solder and solder/Cu electrode interfaces. The phase compositions of IMCs in both EN and ENEPIG were analogous, except for the presence of Pd element as impurity into $Cu_6Sn_5$ phase for ENEPIG. After heating, Pd content in IMCs reduced, and most Pd elements dissolved into solder alloys. In addition, the ENEPIG layer showed excellent wettability with solder, whereas the EN layer presented neutral wetting with solder. The wettability of ENEPIG coating with solder surpassed that of EN, resulting in fewer defect sites at $Bi_2Te_3$/solder interface.

The performance of $Bi_2Te_3$-based TE modules employing EN and ENEPIG diffusion barriers was examined. When current flowed through the modules, EN-coated module exhibited temperature on both hot and cold sides higher than ENEPIG-coated module at most current intensities. While a temperature gradient was applied between two sides of a module, the output power of the module achieved by the ENEPIG layer was higher than that by EN regardless of the temperature gradient. ENEPIG-coated module yielded a lower internal resistance due to its lower contact resistance than EN-coated module. The lower contact resistance achieved by the ENEPIG deposit was attributable to the better wettability of the ENEPIG layer with solder than that of EN coating. Owing to lower internal resistance, ENEPIG-coated module exhibited not only greater output power for power generating performance but also lower temperature on both hot and cold sides for cooling performance than EN-coated module.

After the modules were heat-treated at 200 °C for 200 h, the output power generated by EN- and ENEPIG-coated modules diminished severely. ENEPIG-coated module yielded a marginally higher power than EN-coated module when the temperature gradient was greater than 100 °C. In addition, internal resistance and contact resistance for EN and ENEPIG escalated and the values relating to ENEPIG were less than those relating to EN. As contact resistance was dominant over the resistance of TE legs, it was arguable that the thermal stability of the module after annealing was vastly influenced by the interfacial stability between $Bi_2Te_3$ materials and Cu electrodes.

## CRediT authorship contribution statement

**Yen Ngoc Nguyen:** Validation, Formal analysis, Investigation, Resources, Data curation, Software, Visualization, Writing – original draft. **Kyung-Tae Kim:** Validation, Resources. **Soo-Ho Chung:** Validation, Resources**. Injoon Son:** Investigation, Visualization, Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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